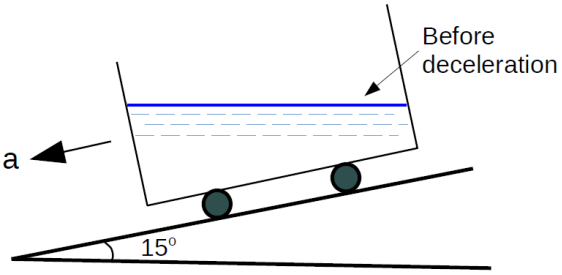
2020-2021 SPRING SEMESTER FLUID MECHANICS MIDTERM EXAM, Date:26/04/2021

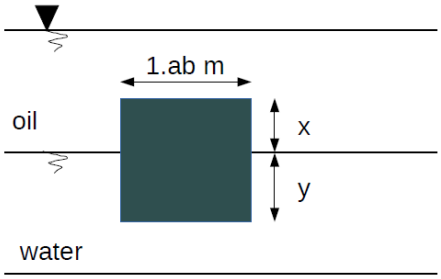
Name: **Surname:** **Student ID:**

Question 4(20p):

a) A tank filled with water is moving down an inclined surface with an angle of 15^o as shown in the figure. If the tank decelerates (slows down) at a= 3.ab m/s^2, determine the angle of the free surface relative to the horizontal direction after deceleration. Draw the water surface after deceleration.



b) A cubical object with a density of d = 0.92a floats at an oil-water interface as shown in the figure. Calculate the distances x and y. (Density of oil is 870 kg/m^3)



a, and b are the last two digits of your student ID.
For example, Student ID: 17042082 a=8 b=2